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Concept Review

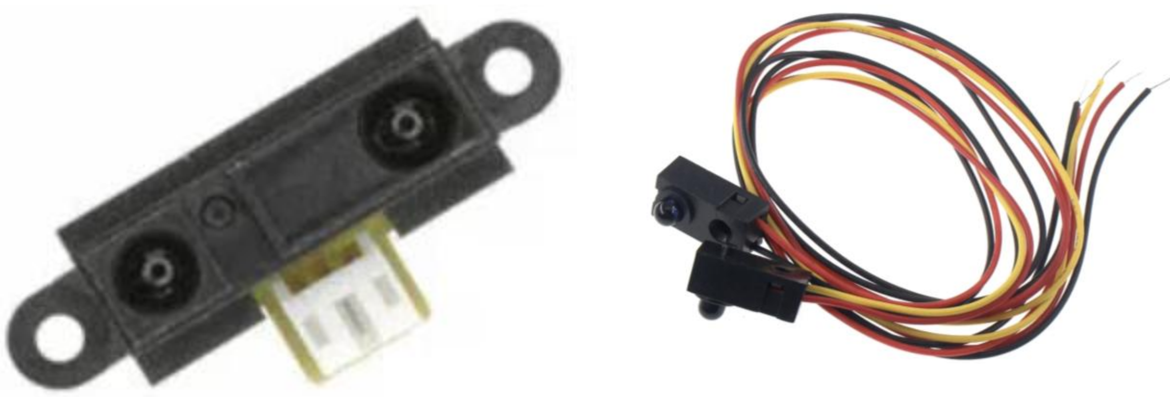
Infrared Distance

Why Do We Need Infrared Distance Sensors?

Distance sensors enable functionality such as proximity sensing, touchless activation, and obstacle avoidance. Infrared distance sensors emit infrared pulses and measure the reflected infrared energy. Infrared distance sensors can be found in applications such as robotics, aviation, and smartphones.

Infrared Distance

Infrared (IR) distance sensors measure distance using the same operating principle as other distance sensors: through emitting and receiving a signal that is reflected by an object. IR distance sensors can be constructed in varying sizes, and two models are shown in Figure 1.



Sharp GP2Y0A21YK0F

Infrared Pair Tube

Figure 1: Different configurations for infrared distance sensors.

An IR distance sensor operates by emitting IR light through an LED or diode and measuring the angle of the reflected light beam using a position sensitive detector. The distance of the object can be estimated by the process of triangulation using the geometry of the reflected light beam. The position of the light beam depends on the distance of the object from the sensor, as shown in Figure 2.

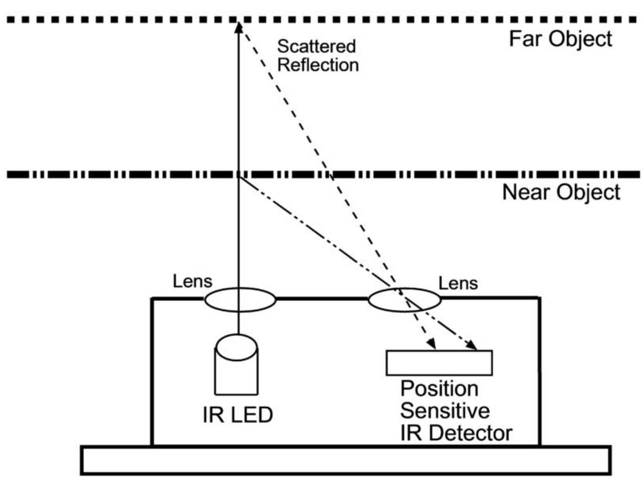


Figure 2: The angle of the reflected light beam is determined by the object distance.

Sound focused distance sensors, such as ultrasonic sensors, use elements such as acoustic cones to focus sound. IR distance sensors operate similarly but instead use lenses to focus the reflected infrared wave.

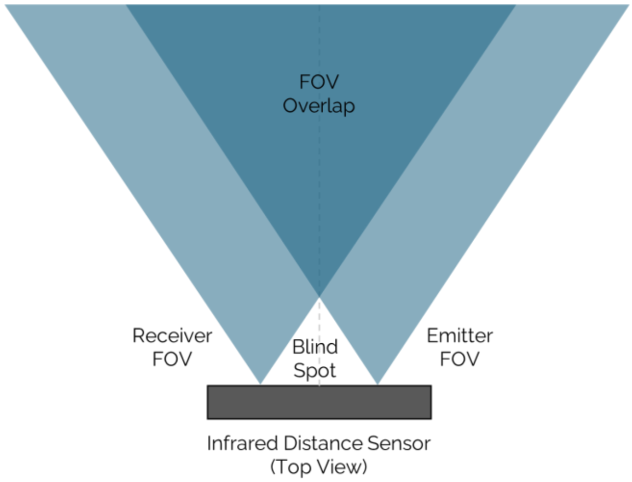


Figure 3: Lens field of view overlap.

As IR distance sensors rely on the use of lenses to focus pulsated infrared waves, they have a fixed limited working range and a blind spot where it cannot reliably detect objects, shown in Figure 3. Like other sensors that contain both an emitter and a receiver, the field of view is determined by the field of view of both components. As an example, for the Sharp GP2Y0A21YK0F the valid distance measurement range is from 10cm up to 80 cm. Another important characteristic of infrared distance sensors is the effect of infrared energy dissipation as a function of distance. As an example, Figure 4. shows the relationship between distance to a reflective object vs Output Voltage for the Sharp GP2Y0A21YK0F.

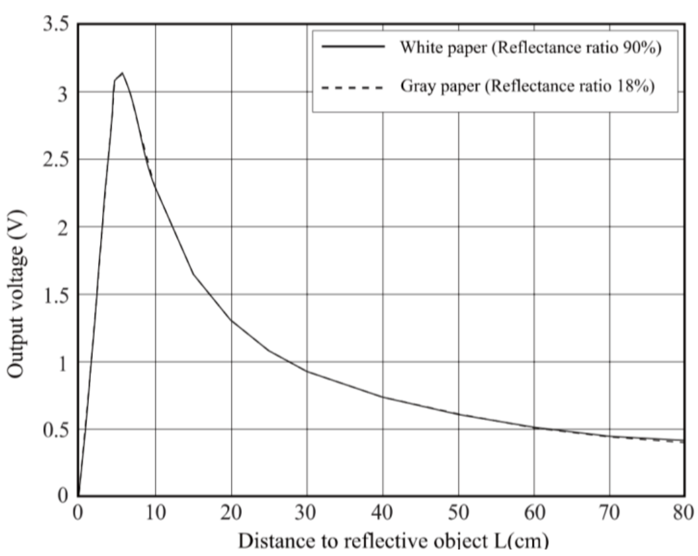


Figure 4: Distance to reflective object vs output voltage

Datasheet and Considerations

Datasheets are used to select an appropriate sensor for a specific application and to determine additional design constraints on the rest of the system. For IR distance sensors, some main datasheet considerations include the following:

Measurement range: This is the distance that the sensor can accurately measure.

Sampling rate or sampling time: This indicates how quickly the sensor can output measurements. The minimum sampling rate depends on the response time requirements for the specific application.

Comparison to Other Distance Sensors

The IR distance sensor falls within a category of sensors that leverage various types of signals for distance measurement. Selecting a distance sensor involves considering several trade-offs between accuracy, measuring range, and cost. Like time-of-flight sensors, which also rely on light, the accuracy of the sensor reading can vary with the reflectivity of surfaces. Both light-based sensors also have shorter response times than the sound-based ultrasonic sensor. Between the two light-based sensors, IR distance sensors measure shorter ranges because they rely on the principle of triangulation instead of time-of-flight.

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